Wayfarer — Trip Planner & Travel Companion

Master task brief · Product spec v1.0 · Platforms: iOS & Android

1. Overview

Wayfarer keeps everything about a trip in one place: flights, hotels, day-by-day itinerary, offline maps, budget and packing list. Booking confirmations are imported automatically from forwarded e-mails. Trips can be shared with travel companions who can edit the plan together. During the trip the app works fully offline and sends timely alerts (gate changes, check-in time, leave-now reminders).

2. Goals

- Create a complete multi-city itinerary in under 10 minutes.
- Import 90% of common booking e-mails (airlines, Booking.com, Airbnb, rail) without manual fixes.
- Everything needed on the road (itinerary, tickets, maps of the planned areas) is available offline.
- Group trips: up to 10 members editing one trip with live updates and a shared expense split.

3. Features

- **Itinerary builder:** days with time slots; add places from search, map or saved lists; drag to reorder; travel time between stops is calculated automatically.
- **Booking import:** forward e-mails to a personal address (trips@wayfarer.app); a parser extracts reservations into the trip.
- **Offline maps:** download map regions for each city in the trip; saved places are pinned.
- **Budget & expenses:** budget per category, expenses in any currency converted to the home currency with daily rates; split between group members and settle up.
- **Packing list:** templates by trip type and weather forecast at the destination.
- **Alerts:** flight status changes, check-in reminders, “leave now” based on traffic.

4. Constraints

Area	Constraint
Privacy	Forwarded e-mails are parsed and deleted within 24 h; only extracted reservation data is kept.
Offline	All trip data stored locally (SQLite); edits made offline are merged on reconnect without data loss.
Cost	Map tiles and flight-status API calls must stay under the budget of $0.05 per active user per month.

5. User flow

From the planning workshop (whiteboard photo). The flow covers the time before and during the trip.

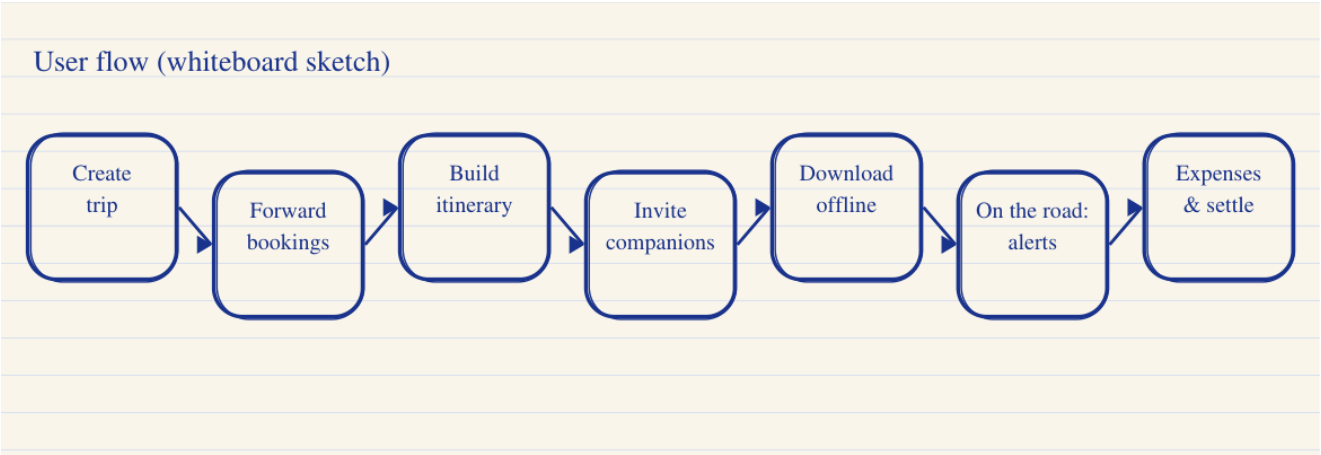


Figure 1. Trip lifecycle user flow (workshop whiteboard).

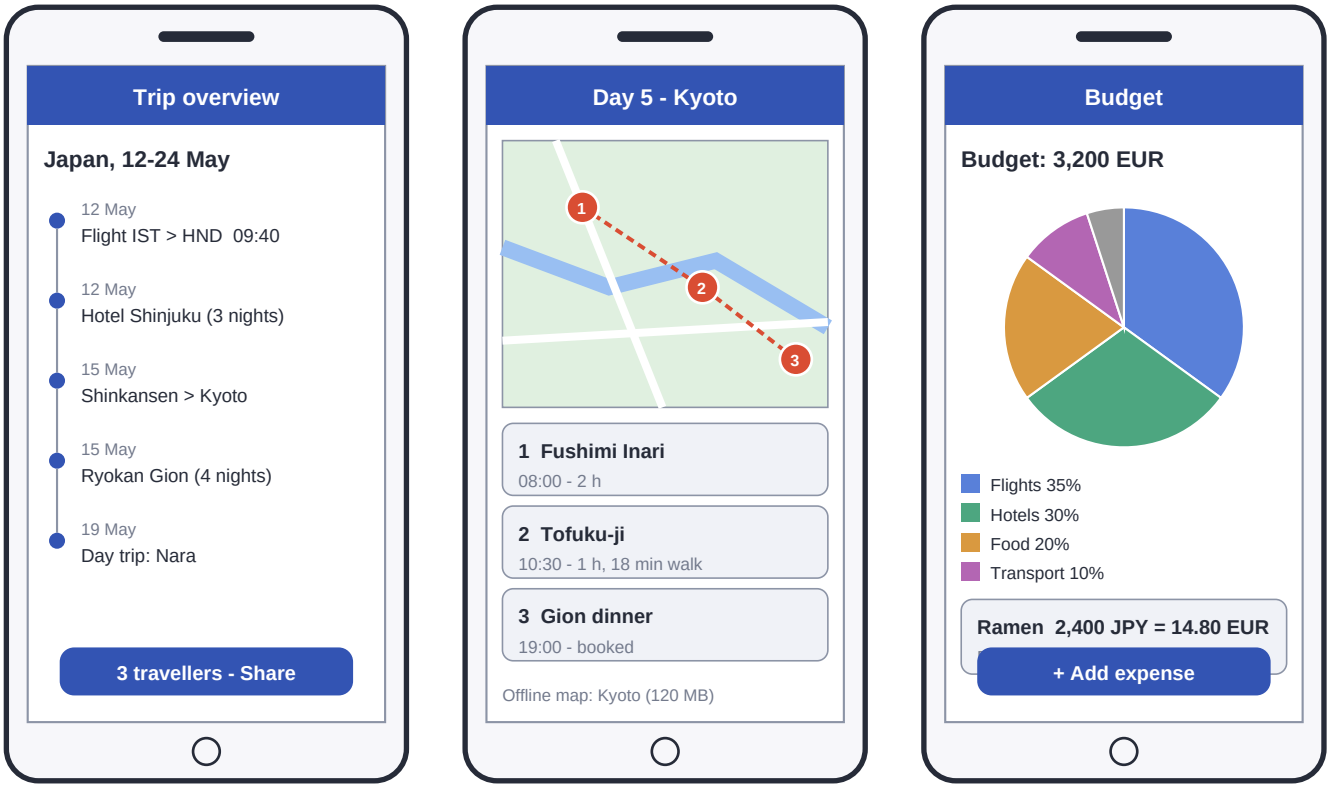


Figure 2. Wireframes: trip timeline, day plan on offline map, budget with currency conversion and split.

6. Notifications

- Flight status from a flight-data API, polled every 15 min from 24 h before departure; push on gate/time change.
- “Leave now” reminder using live travel time to the next booked item.

7. Architecture

Kotlin Multiplatform shared core with native UIs. A sync service keeps group trips consistent using WebSockets and a conflict-free merge for offline edits.

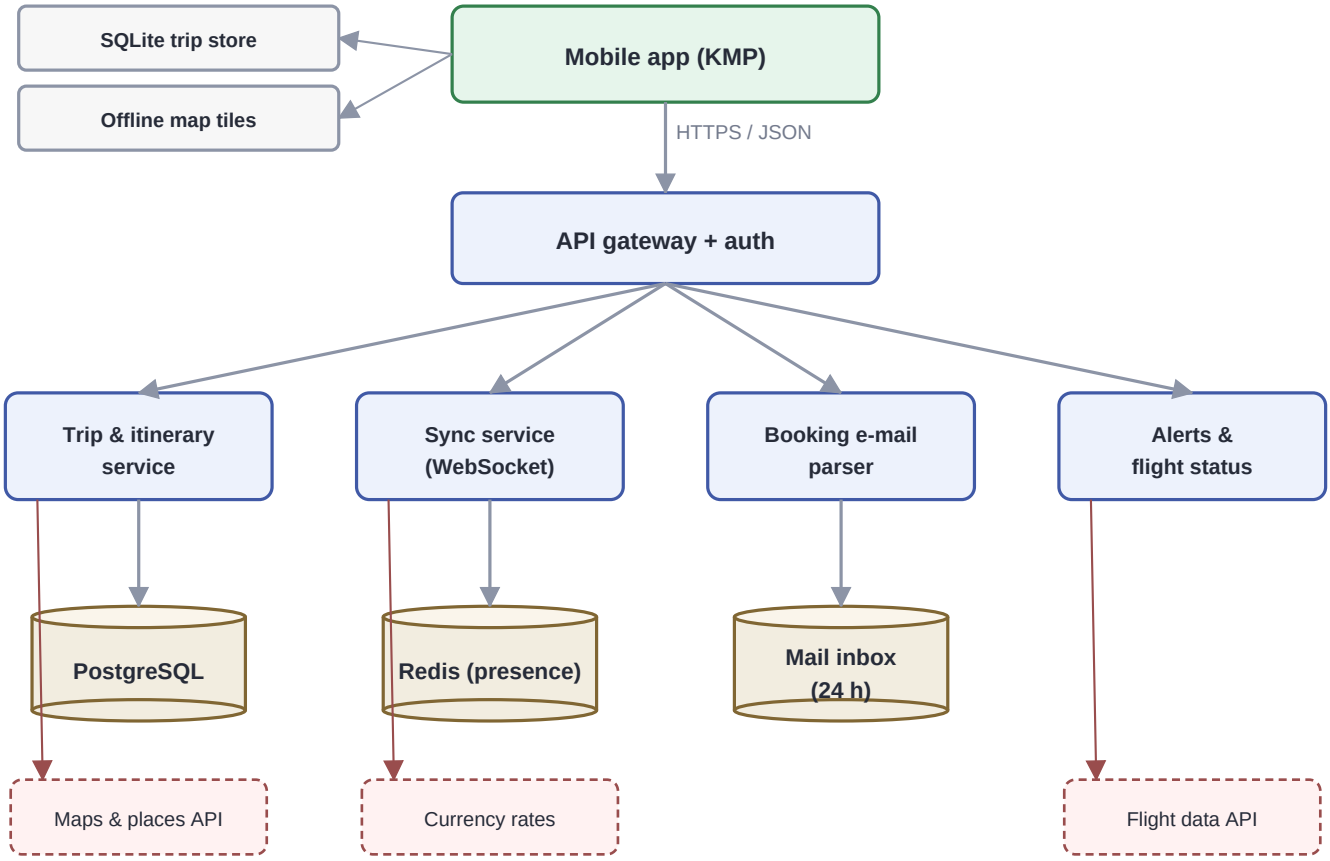
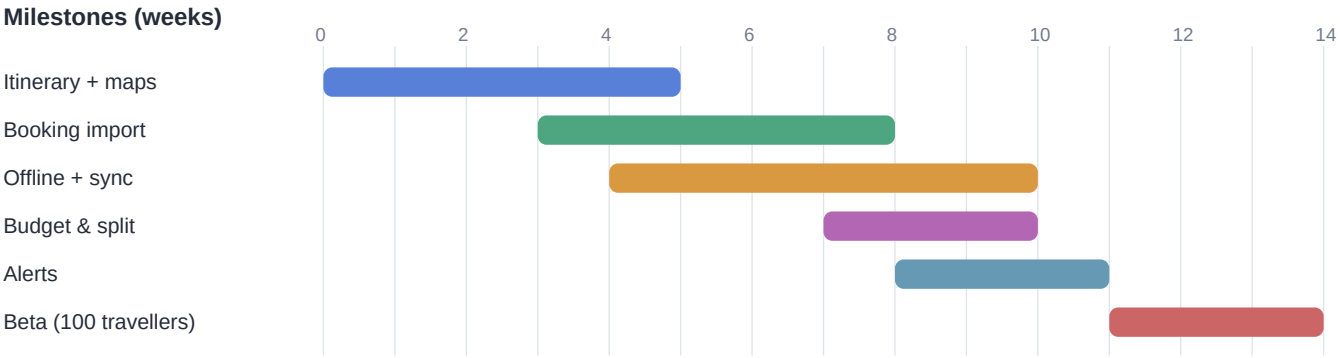


Figure 3. System architecture.



8. Acceptance criteria for v1

- A 2-week, 3-city trip can be used end-to-end in airplane mode after one download.
- Two travellers editing the same day offline see a merged plan after reconnecting, with no lost items.
- Expenses in 3 currencies settle correctly in the home currency (rounding error < 0.01).